

Ning Liang

ning.liang@duke.edu | linkedin.com/in/ning-liang-960621220/ | Phone: +18058699571

Education

Ph.D. Student in Computer Science, Duke University

Durham, NC, USA

Ph.D. Advisor: Dr. Lisa Wu Wills

Aug 2024 - Present

o **Research:**

- * Research focused on domain-specific accelerator, software-hardware codesign, vector database, and hardware simulator.
- * Working on alleviating the bandwidth bottlenecks of large-scale LLM serving via approximation and compression techniques to sustain high-throughput deployments.
- * Interested in pursuing techniques that further improve performance and power efficiency across a wide range of domain-specific applications.

M.S. in Electrical and Computer Engineering, University of Michigan

Ann Arbor, MI, USA

Master's Advisor: Dr. Reetuparna Das, GPA: 4.0/4.0

Aug 2022 - May 2024

M.S. in Data Science, University of Michigan

Ann Arbor, MI, USA

GPA: 4.0/4.0

Aug 2022 - May 2024

B.S. in Mathematics, University of Michigan

Ann Arbor, MI, USA

Double Major in Data Science, GPA: 3.6/4.0

Aug 2017 - Jul 2022

Publications

Jiaao Ma, **Ning Liang**, Ceyu Xu, Alexander van Lanschot, Yunming Xiao, Shihan Lin, Xiaowei Yang, and Lisa Wu Wills. **PhantomBox: Fully Homomorphic Web Application Firewall**. To appear in the International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2027.

Chris Kjellqvist, Mansi Chowdhary, **Ning Liang**, and Lisa Wu Wills. **TinyAct: Activation Functions are (Almost) Linear in Floating Point Representation**. To appear in the International Symposium on Microarchitecture (MICRO), 2026.

Jiaao Ma, Ceyu Xu, **Ning Liang**, and Lisa Wu Wills. **FlashTFHE: A Scalable Architecture for Efficient Multi-bit Fully Homomorphic Encryption**. Proceedings of the 53rd ACM/IEEE Annual International Symposium on Computer Architecture (ISCA), 1909–1923, 2026. [\[paper\]](#).

Ning Liang, Fabian Wenz, Jana Giceva, and Lisa Wu Wills. **Athena: A Plug-and-Play Advisor for Retrieval-Augmented Generation using VectorDB**. 2025 IEEE International Symposium on Workload Characterization (IISWC), 28–41, 2025. [\[paper\]](#).

Yufeng Gu, Alireza Khadem, Sumanth Umesh, **Ning Liang**, Xavier Servot, Onur Mutlu, Ravi Iyer, and Reetuparna Das. **PIM Is All You Need: A CXL-Enabled GPU-Free System for Large Language Model Inference**. Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2, 862–881, 2025. [\[paper\]](#).

Jixiang Yu, Nanjun Chen, Zetian Zheng, Ming Gao, **Ning Liang**, and Ka-Chun Wong. **Chromothripsis detection with multiple myeloma patients based on deep graph learning**. Bioinformatics, 39(7), btad422, 2023. [\[paper\]](#).

Posters

Patrick Hardison, Chris Kjellqvist, **Ning Liang**, and Lisa Wu Wills, “**CoBloom: An FPGA Accelerator System for Bloom Filter Insertion in Genomics Applications**”. IEEE International Symposium for Workload Characterization (IISWC) 2025

Research Experience

PhantomBox: Fully Homomorphic Web Application Firewall

APEX Lab, Duke University

Aug 2025 - Present

- Co-designed an **end-to-end FHE-based WAF datapath** spanning client-side encryption, middlebox rule compilation, and encrypted RegEx execution.
- Contributed to an **ahead-of-time RegEx compiler** with FHE-specific optimizations and equality saturation.
- Developed **hardware-accelerated transciphering** that achieved a **76× speedup over CPU** while keeping on-wire size comparable to plaintext; evaluated 5,800+ attack signatures with **7.7 ms median latency** on AWS EC2 F2.

Related paper: PhantomBox (ASPLOS 2027, to appear) [\[citation\]](#)

- KV Cache Storage Engine for Large-Scale Long-Context LLM Serving**

APEX Lab, Duke University
Aug 2025 - Present

 - Developing and implementing a **disk-friendly, FPGA-accelerated KV cache storage engine** for long-context LLM serving.
 - Investigating **low-bit KV cache compression** targeting **4-bit precision or lower** with substantially reduced accuracy degradation.
- High-Density Special Function Unit (SFU) for AI Accelerators**

APEX Lab, Duke University
Aug 2025 - Present

 - Participated in the design and evaluation of **TinyAct**, a floating-point activation unit that computes exponentiation with a **single integer multiply-accumulate**; SiLU and GeLU require only one additional floating-point FMA.
 - Conducted **post-PNR power, performance, and area evaluation** in TSMC N16, showing **5.19× lower area** than a traditional approximation and **3.4× lower area** than prior approximate hardware.
 - Validated TinyAct on Qwen3, Gemma-3, ViT, and YOLOv8, demonstrating **less than 0.1% inference-quality degradation** and stable LLM training behavior.

Related paper: TinyAct (MICRO 2026, to appear) [\[citation\]](#)
- Characterization of End-to-End RAG System with Vector Database**

APEX Lab, Duke University
Jan 2025 - July 2025

 - Implemented an **end-to-end RAG benchmark suite** covering embedding models, vector databases, approximate nearest-neighbor search, and LLMs.
 - Designed the suite to expose **compute, memory, and data-movement bottlenecks** across the complete RAG pipeline.

Related paper: Athena (IISWC 2025) [\[paper\]](#)
- CXL Memory Expansion for In-Memory Database Systems**

MBit Lab, University of Michigan
Apr 2023 - Apr 2024

 - Developed a **CXL memory-expansion simulator** with an MSI cache-coherence protocol formally verified in Murphi.
 - Validated simulator performance against measurements from **real CXL machines** and performed **instruction-level analysis** of Postgres and Hyrise using Intel Pin.
 - Characterized CXL memory expansion for in-memory databases and developed a **hot-column swap mechanism**.

Submission: Under submission to VLDB 2027.
- CXL-Enabled PIM System for LLM Inference**

MBit Lab, University of Michigan
Apr 2023 - May 2024

 - Co-designed **CXL data-movement instructions** and mapped models up to Llama2-70B across AiM-GDDR6 devices.
 - Explored pipeline, tensor, transformer-block, and hybrid parallelism for a **GPU-free, process-in-memory LLM inference system**.

Related paper: PIM Is All You Need (ASPLOS 2025) [\[paper\]](#)

Internship

- Research Intern**

Ambarella Inc.
May 2026 - Present

Researching state-of-the-art post-training quantization, quantization-aware training, and model pruning methods for vision models, VLMs, and LLMs. Developing a framework to evaluate compression versus accuracy tradeoffs and the fusion of multiple compression methods for deployment on Ambarella's in-house ASIC

Selected Course Projects

- Systolic Array Accelerator for High-dimensional Nearest Neighbor Search**

Duke University
Jan 2025 - Apr 2025

Design a dimension-serial nearest neighbor search accelerator for high-dimensional vectors such as language model embeddings in Chisel HDL, deploy the hardware on Amazon F2 FPGA instances, and achieve more than 3X speedup compared to an A5000 GPU
- Bfloat16 ALU Digital Circuit Design**

Duke University
Sep 2024 - Dec 2024

Designed schematics and layout on transistor level using TSMCN65 library. Minimized transistor count and area with manual place and route. Optimized latency using techniques including carry-look-ahead adder, carry-save multiplier, and barrel shifter
- A Survey of Hardware Acceleration for Quantum Error Correction**

University of Michigan
Jan 2024 - May 2024

Performed a in-depth survey on Quantum ECC decoders including Union-Find, MWPM, Blossom, and QEC accelerators recently published at the time of conducting the survey. Proposed a new architecture based on AFS architecture which supports parity blossom, a fast MWPM decoder
- 3-Way Superscalar Out-of-Order RISC-V Core**

University of Michigan
Sep 2022 - Dec 2022

Implemented an N-Way out-of-order R10K core in Verilog. Built testbenches to test the correctness. Optimized the core with branch prediction, n-way associativity cache, 3-way superscalar, and Store-load forwarding

Language Model with Verifiable Commonsense Language Understanding

University of Michigan

Reproduced results from the paper "Tiered Reasoning for Intuitive Physics: Toward Verifiable Commonsense Language Understanding". Proposed a new transformer based layer to improve model's reasoning ability. Achieved an improvement of 5% in story prediction accuracy and 4% in reasoning verifiability.

Sep 2023 - Dec 2023

1-D U-Net Accelerator Based on Systolic Array

University of Michigan

Implemented, verified, and synthesized a Winograd CNN accelerator in RTL

Sep 2023 - Dec 2023

Skills

Languages: Chisel HDL, Verilog, Python (PyTorch), C/C++, CUDA, MATLAB

Tools: Intel VTune, AMD uProf, Intel Pin, Cadence Virtuoso, Ramulator

Relevant coursework

- Advanced Computer Architecture
- Machine Learning Deep Learning
- GPU Programming
- VLSI and Digital Design
- Database Management Systems
- Quantum Computing Architecture

Teaching

CSE550: Graduate Teaching Assistant

2025

Advanced Computer Architecture

CS250: Graduate Teaching Assistant

2025

Undergraduate Computer Architecture

Achievements and Fellowship

Duke Research Fellowship

2026

Awarded Duke Graduate School Summer Fellowship

Nvidia CUDA Competition, 1st Prize

2023

Achieved the best performance at optimizing a CNN CUDA kernel

James B. Angell Scholars at University of Michigan

2022

Exceptional academic excellence

Dean's Honor at University of California, Santa Barbara

2019

Outstanding academic achievement